

Debasish Chakroborti (Joy), Ph.D.

Computer Science / Software Engineering
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August 2026

Dear Members of the Search Committee,

I am writing to apply for the Assistant Professor position in Computer Science / Software Engineering. I specialize in AI-assisted software maintenance, with a Ph.D. in Computer Science from the University of Saskatchewan (2024), ongoing postdoctoral research, and industry-connected collaborations. I look forward to building an externally funded research program, mentoring graduate and undergraduate students, and contributing to a strong academic unit in software engineering and AI/ML for software systems.

My research aims to deliver issue-free software on the versions people actually use. Under Professors Chanchal K. Roy and Kevin Schneider, I developed Baczer, ReBack, BackSlice, and BackTrans—tools that combine repository mining, machine learning (including CNNs and LLMs), and program analysis to automate software backporting at scale. This work has appeared in venues including ICSE, ICPC, MSR, CASCON, and ASE Journal. Complementary contributions include scientific workflow data reusability (RISP/RISPs) and applied AI automation through my Living Skies Postdoctoral Fellowship (NSERC CREATE SOAR / related Alliance work) and consulting with Cloud Linux Software, Inc. Looking ahead, I plan to advance intelligent release engineering and quality assurance, AI-assisted developer workflows and technical debt, and industry-connected applied software engineering, pursuing funding through opportunities such as NSERC Discovery, Alliance, and Mitacs.

I also bring substantial teaching and mentorship experience that supports a research-active faculty role. I teach in the School of Computing and Digital Innovation at Saskatchewan Polytechnic, served as Sessional Lecturer for CMPT 145 at the University of Saskatchewan with full course responsibility, and previously taught as Lecturer in CSE at Khulna University. I have mentored an M.Sc. student (with co-authored publications), supervised undergraduate summer researchers, and guided teaching assistants. I am prepared to teach core and advanced courses in software engineering, programming, systems, software maintenance, mining software repositories, and AI for software engineering.

I am committed to service and community leadership, including as President of the Computer Science Graduate Council at the University of Saskatchewan, and I have completed Reconciliation Education training. I value collaborative academic cultures and look forward to contributing to departmental service, student success, and equity-minded practices.

Enclosed please find my curriculum vitae, research statement, and teaching statement. I would welcome the opportunity to discuss how my research program and teaching experience can support your department's mission. Thank you for your consideration.

Sincerely,

Debasish Chakroborti (Joy), Ph.D.